

Effect of Tomato Industrial Processing (Different Hybrids, Paste, and Pomace) on Inhibition of Platelet Function In Vitro, Ex Vivo, and In Vivo

Cardiovascular disease (CVD) is the leading cause of death worldwide. Healthy eating is among its safeguards, especially the daily intake of fruits and vegetables. In this context it has been shown that tomato (*Solanum lycopersicum*) presents antiplatelet activity. In the present study, we evaluated in vitro antiplatelet activity of fresh hybrid tomato process (nine hybrids: Apt 410, H 9888, Bos 8066, Sun 6366, AB3, HMX 7883, H 9665, H 7709, and H 9997), paste and its by-product of industrial processes (pomace). We assessed antiplatelet activity ex vivo and bleeding time in rats that ingested 0.1 and 1.0 g/kg of pomace each day. In studies in vitro, no significant differences in antiplatelet activity was observed in fresh tomato hybrids. Furthermore, the agro-industrial process did not affect the antiplatelet activity of paste and pomace. Likewise, pomace intake of 1.0 g/kg per day prolonged bleeding time and reduced ex vivo platelet aggregation in rats. The data obtained indicate that tomato has one or more compounds that caused antiplatelet activity. Regular consumption of tomato and its industrial derivatives could be part of a CVD prevention regimen.